

arttech: Performance and Embodiment in Technology for Resilience and Mental Health

Antje Budde*
University of Toronto
a.budde@utoronto.ca

Mark Chignell
University of Toronto
chignell@mie.utoronto.ca

Jamy Li
Ryerson University
jamy@ryerson.ca

ABSTRACT

This workshop explores socially expressive technology design using collaborative principles of theater, dance and music. Building on shared roots of making something together, technologies (including virtual agents, physical devices, storytelling multimedia) and artistic provocation can create playful, productive and healthy experiences for communities. How do we: Improve collaboration between performance art and creative technology? Design apps and artistic collaboration/learning for mental health? Use theater performativity and technology to enhance our lives? As smart devices become ubiquitous, yet solidarity and cooperation remain elusive in competitive/hierarchical societies, understanding how performance art and technology collaborate can improve how we work, play and care for our and each other's health. This workshop is a forum and creative space to discuss and play with the future of performative expression and collaborative computing. We welcome researchers and practitioners from varied disciplines including technical implementation, design, theater, dance, improv, physiology and psychology. The program features collaborative creation modules across the expressive human spectrum (movement, sound, text) using digital dramaturgy (Budde, 2017) and pleasure of learning (Brecht, 2001). It also includes participants' presentations, expert talks and videos exploring the expressive side of computing technology. Accepted applicants are invited to share perspectives and collaborate on creative exercises on art/artefacts/artificiality.

CCS CONCEPTS

• **Applied computing**; • **Performing arts**; • **Human-centered computing**; • **Collaborative and social computing design and evaluation methods**; • **Consumer health**;

KEYWORDS

Mental Health, VR/AR, Game Design, Embodiment, Performance

ACM Reference Format:

Antje Budde, Mark Chignell, and Jamy Li. 2021. arttech: Performance and Embodiment in Technology for Resilience and Mental Health. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '21 Companion)*, October 23–27, 2021, Virtual Event, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3462204.3481728>

*authors listed in alphabetical order

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW '21 Companion, October 23–27, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8479-7/21/10.

<https://doi.org/10.1145/3462204.3481728>

1 INTRODUCTION AND BACKGROUND

Symptoms of student depression, anxiety, eating disorders, and other mental illnesses have risen, along with demand for counseling services [1, 2]. Universities, for example, face challenges when attempting to prevent, identify, and treat mental illness on campus [3] (p. 767), and social media may exacerbate the challenges students face [4]. Many students may experience loneliness, anxiety and depressed mood as they deal with the pressure of rapidly learning new ideas and skills in a competitive university environment and dealing with a confusing new world of social relationships and ideologies. Other special populations such as older adults, particularly those who are also minorities, likewise face growing challenges with loneliness [5]. This workshop focuses on advancing the wellness and self-learning (including resilience, mental health and alleviation of loneliness) of diverse groups of society. It aims to expand research on how technologies can create opportunities for personal growth and enhancement of resilience through embodied and meaningful interactions with multimedia content, and with others. The activities and challenges that we showcase will involve alternative thinking (play, imagination, problem solving) such as creative expression (cf. [6] and performance-based analysis (cf. [7]) that are integrated with embodiment and the senses.

The Interactive Media Lab's previous work has shown the value of using exergaming to improve cognitive and physical health (e.g., [8, 9]). The Ryerson Tech Lab's previous work has included theater-inspired robotics (e.g., [10]). The Digital Dramaturgy Lab^{squared} has shown how dramaturgy can facilitate playful learning, creative exploration, and building community in a warm and safe environment [6, 11] (see Figures 1-3). Building upon our existing dramaturgical methodology and exergaming framework, we aim to discuss novel methods to provide self-learning and insight, and to thereby increase resilience, for students and other special populations who are struggling to cope with the stresses of daily life. Our focus will be on promoting well-being rather than tackling pathology and on exploiting the benefits of technology rather than being technology driven. This research can benefit from a rich ecosystem for supporting wellness initiatives (through CSCW research in wellbeing, distributed teams, multidisciplinary inquiry) and technology development (such as augmented/virtual reality, robotics and conversational user interfaces).

arttech@CSCW2021 will employ play and game dramaturgy/action design informed by Brecht's learning play [12, 13], where the primary goal of performances is for the actors (our workshop participants) to acquire critical attitudes/skills (rather than produce entertainment). As part of the workshop activities, we will present motion-prompted scenarios/interfaces, experimenting with case studies (e.g., fear and anxieties, perfectionism freezes, feelings of isolation, loneliness/imposter syndrome,



Figure 1: Left: Jing@Ju Karaoke – Interactive Queering of Beijing Opera (2015). Right: From my (global) village I see as much— Walking inside a poem/ Multi-media explorative performance installation inspired by Fernando Pessoa in Toronto (2014). Photo credit: Antje Budde



Figure 2: Stare. Print. Blue. – Voyeurizing the Apparatus (2013, 2014). Photo credit: Antje Budde



Figure 3: Artaud's Cage (2013). Photo credit: Antje Budde

navigation of personal and professional challenges). This is relevant to CSCW researchers interested in the collaborative creation process, play and communication through embodied design.

arttech@CSCW2021 aligns with past workshops in healthtech (e.g., [14] at CSCW2015), arttech (e.g., [15] at CHI2017) and human-technology interaction workshops (e.g., [16]), though we believe

our online-adapted collaborative activities and focus on resilience is particularly important given the growing problems of activating interaction within virtual meeting spaces and of addressing mental health.

2 CALL FOR PARTICIPATION: ARTTECH@CSCW2021

Virtual (dis)connection and mental health challenges are growing issues faced during a global pandemic. The overall goal of this workshop is to bring together researchers and practitioners interested in different scenarios for interacting with wellbeing technology and the roles that the performing arts and collaborative technology can play for social purposes like mental health, education, play, and workplace life. In order to maximize our chances of success, workshop participants will present scholarship, join collaborative activities and engage with Canadian and international artists and scientists. First, we will bring together scholarship that advances the practice and knowledge of wellbeing technology.

This can include (but is not limited to):

- mental health technologies (VR, robots, voice agents, apps, etc.)
- embodied self-learning strategies that can build psychological resilience
- art- or theater-inspired design
- mental health of distributed teams
- designing art-related technology for users from special populations
- theater-inspired ways to improve mental and physical health
- expressive 'probes' for learning, mental health, dining, home life, etc.

Second, this workshop will host a series of online collaborative creation modules that offer participants a collaborative opportunity to: playfully hack app functionality; practice improvisational and alternative thinking scenarios; raise critical awareness of human-machine interFACES; practice playful togetherness; and think about applications of performative techniques and digital dramaturgy in research and innovation. Third, this workshop features both invited artists and invited participant-speakers who will share methods and scholarship.

3 PRE-WORKSHOP SELECTION OF PARTICIPANTS

Applicants are asked to submit a position paper (2 – 4 pages in ACM submissions template; please see Step 1 of “The Workflow and Templates” found here: <https://www.acm.org/publications/taps/word-template-workflow>) on their ongoing projects centered on arttech, including work-in-progress and late-breaking results. The organizing committee will review the position papers and accepted papers will be presented by participants during the participant contributions session (see Workshop Dates, Format and Schedule). We expect 10-20 participants.

For potential applicants who have limited funds, the workshop organizers will be providing need-based scholarships, which cover the cost of one CSCW conference registration and one arttech@CSCW2021 registration.

3.1 Invited Participants

arttech@CSCW2021 will feature two categories of invited participants in addition to accepted applicants: 3 invited artists and 4 invited participant-speakers.

Invited artists will play a key role in facilitating collaborative activities. The organizers will privately invite young artists from within our network (TBA) and fund their conference/workshop fees as well as an honorarium. Invited participant-speakers will give presentations during the participant contributions. Invited participant-speakers include Eleanor Dare (confirmed) and Matt Lewis (confirmed), who are affiliated with the Royal College of Art, as well as James Young (confirmed) from Computer Science at University of Manitoba and Frauke Zeller (confirmed) from Professional Communication at Ryerson University.

4 WORKSHOP DETAILS

arttech@CSCW2021 is a two-day online workshop consisting of one session of participant contributions (i.e., presentations, including invited talks; see above) and three sessions of collaborative activities.

4.1 Collaborative Activity Themes and Goals

The themes for the collaborative activities held during arttech@CSCW2021's are mental health, creative collaboration, alternative thinking, digital dramaturgy, pleasure of learning. The underlying principles that guide the activities include: Radical slowness (an emphasis on process rather than an end product); togetherness/non-exploitative collaboration (in contrast to the negative effects of competition, e.g., [17]); artistic intelligence (or “A/I”, as a contrast to artificial intelligence); laughter/critical doubt (not taking oneself too seriously) [6]. These are related to the principles of critical making ([18] and the Critical Making Lab, Toronto), collaborative aesthetic [19] and cooperative aesthetic (Funk and the University of Art and Design Linz; cf. [20]), and theatrical machines [21].

4.2 Collaborative Activity Descriptions

In a virtual meeting we will not be able to use the physical interfaces that were proposed for our PLAYStrong workshop series (see Acknowledgments). Nevertheless, we will select breakout-room-friendly exercises such as:

- Creative writing/sounding performance: Playing with MS Word online: text, structure, voice and multi-linguality. Creating a stream of consciousness poetic text, read by build-in voice options.
- Visual storytelling performance: Playing with making images/imagination with MS PowerPoint. Inspired by Bertolt Brecht's model of epic theatre, the “*Street Scene*” [7], tell a visual story of street scenes as they occur in the moment of collaboration using digital photography and spontaneous voice-over
- Animating move/movie performance: Playing with illusion techniques of movement using MS PowerPoint like a flipbook animation tool. Inspired by the surrealist drawing game *exquisite corpse* (<https://www.artsy.net/article/artsy-editorial-explaining-exquisite-corpse-surrealist-drawing-game-die>)
- Digital puppetry/mask/noise performance: Playing with facial expressions that animate digital puppets or masks using the SnapCamera application to create a real-time improvised dialogue with collaborating performers/animators. Inspired

Table 1: Workshop schedule with activities and times in three sample time zones.

Day Duration	Activity (see Section 4)	Start time in EDT (US)	Start time in BST (UK)	Start time in CST (CN)
Day #1: Oct 23				
0.5 h	introduction of themes and methods	9:00 AM	2:00 PM	9:00 PM
1.5 h	participant contributions, incl. invited talks	9:30 AM	2:30 PM	9:30 PM
0.5 h	break	11:00 AM	4:00 PM	11:00 PM
1.5 h	collaborative activities #1	11:30 AM (1:00 PM end)	4:30 PM (6:00 PM end)	11:30 PM (1:00 AM * end)
Day #2: Oct 24				
0.5 h	debrief and discussion of previous day	9:00 AM	2:00 PM	9:00 PM
1.0 h	collaborative activities #2	9:30 AM	2:30 PM	9:30 PM
0.5 h	break	10:30 AM	3:30 PM	10:30 PM
1.5 h	collaborative activities #3	11:00 AM	4:00 PM	11:00 PM
0.5 h	reflection	12:30 PM (1:00 PM end)	5:30 PM (6:00 PM end)	12:30 AM * (1:00 AM * end)

by the worldwide tradition of mask and puppet animation that goes back thousands of years, translating such praxis into the current moment.

4.3 Workshop Dates, Format and Schedule

The workshop dates are Sat.-Sun. Oct. 23-24, 2021 (4h per day incl. 30 min break, see Table 1).

5 POST-WORKSHOP KNOWLEDGE DISSEMINATION

We will disseminate the workshop schedule, materials (e.g., videos related to the collaborative activities and principles covered in this workshop) and participants on a dedicated website: <https://arttechworkshop.square.site>.

We are in discussion with Morgan Claypool to publish a book based on our research, this workshop experience and a review of associated literature. We expect that the book will be published in the series: Synthesis Lectures on Assistive, Rehabilitative, and Health-Preserving Technologies, edited by Ron Baecker and Andrew Sixsmith. We will acknowledge CSCW in the book and we will also reference the research of the workshop participants.

6 ORGANIZERS

Antje Budde (Drama, Theatre and Performance Studies) is a queer-feminist, multi-lingual scholar-artist of applied theatre sciences and cultural communication. She is an expert in digital dramaturgy as experimental performance, human-machine agency in live performance and practices of participatory learning plays through scenarios of alternative thinking. **Mark Chignell** (Industrial Engineering) is an expert in human factors and information engineering with a particular interest in health applications. His goal is to improve quality of life across the lifespan. This collaborative research will create a synergistic merging of our approaches to interactive media performances in support of skill building for sustainable mental health in university students and other populations. **Jamy**

Li (Industrial Engineering) is a human-robot interaction researcher with an interest in special populations.

ACKNOWLEDGMENTS

Antje Budde and Mark Chignell would like to acknowledge that their part of this work has been funded by a University of Toronto XSeed grant (2021-2023) entitled: “PLAYStrong – Promoting Student Resilience through Interactive Prototypes for Embodied Self-learning”. Jamy Li’s participation in the workshop and associated research is funded by an NSERC Discovery Grant (2021-2026)

REFERENCES

- [1] Lattie, E. G., Lipson, S. K., & Eisenberg, D. (2019). Technology and college student mental health: challenges and opportunities. *Frontiers in Psychiatry*, 10, 246.
- [2] Lipson, S. K., Lattie, E. G., & Eisenberg, D. (2018). Increased rates of mental health service utilization by US college students: 10-year population-level trends (2007–2017). *Psychiatric Services*, 70(1), 60–63.
- [3] Jaworska, N., DeSomma, E., Fonseka, B., Heck, E., & MacQueen, G. (2016). Mental health services for students at post-secondary institutions: A national survey. *Canadian Journal of Psychiatry*, 61(12), 766–775.
- [4] Elhai, J. D., Dvorak, R. D., Levine, J. C., & Hall, B. J. (2017). Problematic smartphone use: A conceptual overview and systematic review of relations with anxiety and depression psychopathology. *Journal of Affective Disorders*, 207(2017), 251–259. <https://doi.org/10.1016/j.jad.2016.08.030>
- [5] Gierveld, J. D. J., Van der Pas, S., & Keating, N. (2015). Loneliness of older immigrant groups in Canada: Effects of ethnic-cultural background. *Journal of Cross-Cultural Gerontology*, 30(3), 251–268.
- [6] Budde, A. with Babayants, A. (2020). *Homo Ludens – Common. Play. Grounds. A praxis report on workshop-making by the Digital Dramaturgy Labsquared (DDL2) at Coburg University of Applied Sciences, Germany 2019*. In J. Heinicke & K. Lohbeck (Eds.), *Elfenbeinturm oder Kultur für Alle? Kulturpolitische Perspektiven und künstlerische Formate zwischen Kulturinstitutionen und Kultureller Bildung* (pp. 111–143). KoPaed Publishing, Munich.
- [7] Brecht, B. (2001) 2.2. Epic Theatre. Bertolt Brecht from “The Street Scene”. In C. Counsell & L. Wolf (Eds.), *Performance Analysis: An Introductory Coursebook* (pp. 43–48). Routledge, London.
- [8] Chignell, M., Matulis, H., & Nejati, B. (2020, July). Motivating physical exercise in the elderly with mixed reality experiences. In *International Conference on Human-Computer Interaction* (pp. 505–519). Springer, Cham.
- [9] Tong, T., Chan, J. H., & Chignell, M. (2017, April). Serious games for dementia. In *Proceedings of the 26th international conference on World Wide Web Companion* (pp. 1111–1115).
- [10] Li, J., Cuadra, A., Mok, B., Reeves, B., Kaye, J., & Ju, W. (2019a). Communicating dominance in a nonanthropomorphic robot using locomotion. *ACM Transactions on Human-Robot Interaction (THRI)*, 8(1), 1–14.

- [11] Budde, A. (2017). Affecting the apparatus: Queer feminist re/de-codings in the works of the Digital Dramaturgy Lab (DDL), Toronto. In E. Diamond, D. Varney, & C. Amich (Eds.), *Performance, Feminism and Affect in Neoliberal Times* (Contemporary Performance Interactions Series) (pp. 189-200). Palgrave.
- [12] Pinchbeck, D. (2006, April). A theatre of ethics and interaction? Bertolt Brecht and learning to behave in first-person shooter environments. In *International Conference on Technologies for E-Learning and Digital Entertainment* (pp. 399-408). Springer, Berlin, Heidelberg.
- [13] Mueller, R. (2006). Learning for a new society: The Lehrstück. In P. Thomson & G. Sacks (Eds.), *The Cambridge Companion to Brecht* (Cambridge Companions to Literature) (pp. 101-117). Cambridge: Cambridge University Press. <http://dx.doi.org/10.1017/CCOL0521857090.007>
- [14] Chamberlain, A., Schraefel, M. C., Poole, E., Munson, S., Danis, C., & Churchill, E. (2015, February). Moving beyond e-health and the quantified self: The role of CSCW in collaboration, community and practice for technologically-supported proactive health and wellbeing. In *Proceedings of the 18th ACM Conference Companion on Computer Supported Cooperative Work & Social Computing* (pp. 273-276).
- [15] Fitter, N. T., Knight, H., Martelaro, N., & Sirkin, D. (2017, May). What actors can teach robots. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (pp. 574-580).
- [16] Li, J., Sirkin, D., van Erp, J., & van Riemsdijk, B. (2019b). Closeness with robots as social partners. In *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 691-692). IEEE.
- [17] Harrison, L. K., Denning, S., Easton, H. L., Hall, J. C., Burns, V. E., Ring, C., & Carroll, D. (2001). The effects of competition and competitiveness on cardiovascular activity. *Psychophysiology*, 38(4), 601-606.
- [18] Ratto, M. (2011). Critical making: Conceptual and material studies in technology and social life. *The Information Society: An International Journal*, 27(4), 252-260. <http://dx.doi.org/10.1080/01972243.2011.583819>
- [19] Blakinger, J. R. (2016). The aesthetics of collaboration: Complicity and conversion at MIT's Center for Advanced Visual Studies. *Tate Papers*, 25. <https://www.tate.org.uk/research/publications/tate-papers/25>
- [20] D'Auria, V. (2018, September 17). Cooperative Aesthetics ar *Ars electronica* 2018. AnnaMonteverdi. Available: <https://www.annamonteverdi.it/digital/cooperative-aesthetics-ar-ars-electronica-2018/>
- [21] Butler, J. (2015, Nov). Theatrical machines. *Differences*, 26, no. 3, 23–42. <https://doi.org/10.1215/10407391-3340336>